Rahul Podugu

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Open to relocation within the U.S. (assistance welcomed but not required)

Professional Summary

Software Engineer with over 3 years of experience in Java development, Spring Boot microservices, and cloud-native applications on AWS. Skilled in designing scalable payment systems, implementing AI/ML solutions, and optimizing application performance. Experienced in Agile methodologies, CI/CD pipelines, and containerization with Docker and Kubernetes. Focused on applying AI, microservices architecture, and cloud technologies to build robust enterprise solutions.

Technical Skills

- **Programming Languages:** Java, Python, JavaScript, TypeScript, C++, C#, Kotlin, HTML, CSS
- **Frameworks & Libraries:** Spring Boot, Spring Framework, Node.js, Express, Angular, Angular Universal, React, REST APIs, GraphQL, gRPC, FastAPI
- **Cloud & DevOps:** AWS (EC2, Lambda, S3, EKS, CloudWatch, IAM), Azure, Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, Helm, Prometheus, Grafana, CI/CD, Git
- **Data & AI:** PostgreSQL, MongoDB, Redis, Apache Kafka, Milvus, Pinecone, TensorFlow, scikit-learn, LangChain, OpenAI API, Claude API, LLMs, RAG Pipelines, Vector Databases, Prompt Engineering
- **Software Engineering & Architecture:** Microservices Architecture, Event-Driven Architecture, Saga Pattern, Distributed Systems, API Design, System Design, Performance Optimization, ETL Pipelines, Data Migration
- **Security:** JWT Authentication
- **Testing & Agile:** JMeter, Selenium, JUnit, Mockito, Jest, Postman, TDD/BDD, Agile, Scrum, Kanban, Jira

Professional Experience

- **Aztra** — Aug 2024 – Present
  Software Developer
  - Architected and deployed Java Spring Boot microservices on AWS to support 100,000+ daily transactions, achieving 99.9% uptime for the payment platform.
  - Integrated Apache Kafka to enable an event-driven architecture and implemented the Saga pattern for distributed transactions, reducing end-to-end latency by 40%.
  - Engineered an AI-driven search platform using LangChain (GPT-4) and a Milvus vector index on AWS S3 to surface relevant logs and runbooks for root-cause analysis, cutting mean time to recovery by 40%.
  - Optimized PostgreSQL performance by tuning SQL queries and implementing Redis caching, reducing API response times by 30%.
  - Conducted JMeter load testing and configured AWS Auto Scaling with Elastic Load Balancing, maintaining the 99th-percentile latency under 150ms for 50,000+ concurrent users.
  - Led a 3-person Scrum team and established CI/CD pipelines using GitHub Actions and Jenkins, automating build, test, and deployment workflows.
  - Containerized microservices with Docker and Kubernetes and integrated Prometheus and AWS CloudWatch for monitoring, improving observability and enabling 40% faster incident recovery.
- **University of Maryland, Baltimore County** — Sep 2022 – Nov 2023
  Software Developer
  - Developed a real-time research analytics dashboard with Angular Universal (server-side rendering), improving page load time by 40% and enabling faster data insights for research users.
  - Designed and deployed a Node.js backend (Express + GraphQL) on Kubernetes to support high-concurrency data visualization queries for research teams.
  - Secured sensitive research data with JWT authentication and role-based access control (RBAC), ensuring compliance with university data governance policies.
  - Collaborated with faculty and researchers to turn complex visualization requirements into interactive dashboards using Chart.js and D3.js, increasing data accessibility by 20%.
- **UST** — Sep 2021 – Jun 2022
  Software Developer
  - Designed and implemented resilient Java-based payment orchestration services using Apache Kafka and the Saga pattern, ensuring ACID-like consistency and end-to-end data integrity across microservices.

- Added robust error handling, retry, and recovery logic, improving transaction reliability by 45% for 10,000+ daily payments.
- Refactored Spring Boot services for asynchronous REST communication, boosting system throughput by 30% and reducing inter-service latency.
- Developed robust REST APIs with real-time validation and concurrency control, improving warehouse inventory accuracy by 25%.
- Created test automation suites using Selenium WebDriver, JUnit, and Mockito in a TDD workflow, reducing manual QA effort by 75%.
- Led Agile sprints and Kanban workflows as acting Scrum Lead for a 6-person team, accelerating delivery velocity by 25%.

- **Fidelity National Financial** Apr 2021 – Aug 2021
  Software Engineer
  - Streamlined large-scale ETL pipelines for an Oracle-to-Redshift data migration, resolving 200+ data discrepancies and improving data quality by 50%.
  - Developed a Java-based automation framework for data extraction and transformation, cutting ETL processing time by 30%.
  - Optimized Hibernate queries and refactored Java modules to eliminate memory leaks, reducing system crashes by 40%.
  - Automated end-to-end ETL deployment workflows on AWS with Jenkins and Docker, improving pipeline reliability and eliminating manual configuration.

Projects

FinSight AI *AWS-Native Fintech RAG Platform*
`github.com/rahul2251999/FinSight-AI`

- Architected an end-to-end RAG system on AWS (FastAPI, LlamaIndex, OpenSearch Serverless, S3) to deliver semantic retrieval across payment regulations, KYC/AML policies, and fraud playbooks.
- Fine-tuned and deployed an open-source LLaMA model with QLoRA on Amazon SageMaker for scalable inference, improving answer consistency and reducing hallucinations by 35%.
- Developed a transaction-aware risk engine that fuses RAG outputs with DynamoDB-backed merchant and anomaly data to surface real-time risk scores and remediation insights, running on ECS Fargate with CI/CD and CloudWatch monitoring.

Decentralized File Sharing System *P2P Distributed Systems*
`github.com/Rahul2251999/Peer-to-Peer-Secured-File-System`

- Engineered a peer-to-peer file-sharing system using Java, RSA/AES hybrid encryption, and DHT-based indexing to ensure secure, decentralized content discovery and transfer.
- Implemented a multithreaded networking layer with Linux `io_uring` to optimize I/O throughput and reduce system-level overhead during large file transfers.
- Achieved 3x faster file access and retrieval speeds compared to centralized file-sharing pipelines through concurrency tuning and low-level I/O optimization.

AI Customer Support Chatbot *AI/NLP Automation*
`github.com/rahul2251999/AI-Powered-Customer-Support-Chatbot`

- Built an AI-powered customer service chatbot using Azure Bot Services, LUIS, and a custom NLP pipeline to automate intent classification and response generation.
- Integrated the bot with enterprise knowledge bases and transactional APIs to deliver contextual, multi-turn conversations tailored to support workflows.
- Handled 1,000+ daily customer interactions while improving first-contact resolution rates by 60%.

Education

- **University of Maryland Baltimore County** Aug 2022 – May 2024
  M.S. in Computer Science

Publications

- "Remote Medical Assistance for Marine Fishermen through OceanNet", IEEE ICCCNT 2021, DOI: 10.1109/ICCCNT51525.2021.9579678